CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

BCMA: 2 molecules stopped early

4 stopped trials across 2 molecules: Belantamab Mafodotin and Autologous BCMA/TACI-targeted CAR T Cells AUTO2. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

13.8%
4 of 29 closed trials (95% CI 5.5%–30.6%)

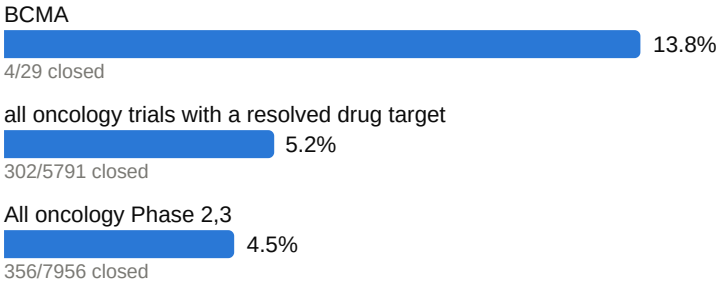
4 / 0 / 0
efficacy / safety / both (adds to 4)

21.6%
of 134 trials have closed · 105 still open or unresolved

2.4%
stopped for a biological reason within 3 years of starting (95% CI 0.6%–9.2%; 67 still at risk; 1.5% area-wide)

— **Completed and missed the primary endpoint.** None of the 17 completed trials in this segment posted a primary result on ClinicalTrials.gov that can be read, so nothing can be said here either way. 2 of the 14 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 68 distinct experimental drugs across 134 trials (antibody 66, small molecule 52, adc 42, gene therapy 22). Most tested: BELANTAMAB MAFODOTIN (41), DEXAMETHASONE (32), LENALIDOMIDE (22), TECLISTAMAB (19), BORTEZOMIB (16). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 4 stops came from 4 sponsor-asset programmes across 3 sponsors; the largest (Jacob Soumerai, MD) contributed 1; removing that programme's trials from both sides leaves 10.7%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 3 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 24.1%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2018–2020	1	10	10.0%	1.8%–40.4%
2021–2024	2	17	11.8%	3.3%–34.3%

Where the stops sit

Efficacy / futility: 4.
Sponsors with most stops: GSK (2), Jacob Soumerai, MD (1), Autolus Limited (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT06655818	2	GSK GSK	BELANTAMAB MAFODOTIN, DOSTARLIMAB	2024-02	Efficacy	The study was terminated due to lack of efficacy
NCT04676360	2	Jacob Soumerai, MD	BELANTAMAB MAFODOTIN	2023-11	Efficacy	The study was closed given lack of efficacy and plans to publish this negative study.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT06160609	2	GSK GSK	BELANTAMAB MAFODOTIN, GSK-3174998	2023-01	Efficacy	The study was terminated due to lack of efficacy.
NCT03287804	2	Autolus Limited AUTL	Autologous BCMA/TACI-targeted CAR…	2019-09	Efficacy	Preliminary efficacy seen to date following treatment with AUTO2 has been determined not sufficient to warrant further development

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 21.6% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.